

Lake Powell and Lake Mead Historical and Current Reservoir Data

David E. Rosenberg

This report was generated on: May 17 , 2026

Overview

This is an R Markdown document. The document generates plots for Lake Powell and Lake Mead:

1. Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

Data wrangling strategy

1. Import the reservoir elevation-volume curves.
2. Import Daily, Monthly, and Annual Lake Powell and Lake Mead data from the Reclamation's HydroData Web portal https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html
3. Filter the reservoir data for the appropriate variables/fields.

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

Requested Citation

David E. Rosenberg (2026), "Lake Powell and Lake Mead Historical and Current Reservoir Data." Utah State University. Logan, Utah. <https://github.com/dzeke/ColoradoRiverCollaborate/tree/main/HistoricalCurrentReservoirData>.

```
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/HistoricalCurrentRe
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/df
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/df
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/df
## [1] "Successfully read - Lake Powell : Storage - https://www.usbr.gov/uc/water/hydrodata/reservoir
## [1] "Successfully read - Lake Powell : Evaporation - https://www.usbr.gov/uc/water/hydrodata/reser
## [1] "Successfully read - Lake Powell : Inflow - https://www.usbr.gov/uc/water/hydrodata/reservoir_
## [1] "Successfully read - Lake Powell : Inflow Volume - https://www.usbr.gov/uc/water/hydrodata/res
## [1] "Successfully read - Lake Powell : Unregulated Inflow - https://www.usbr.gov/uc/water/hydrodat
## [1] "Successfully read - Lake Powell : Unregulated Inflow Volume - https://www.usbr.gov/uc/water/h
## [1] "Successfully read - Lake Powell : Total Release - https://www.usbr.gov/uc/water/hydrodata/res
## [1] "Successfully read - Lake Powell : Release volume - https://www.usbr.gov/uc/water/hydrodata/re
## [1] "Successfully read - Lake Powell : Pool Elevation - https://www.usbr.gov/uc/water/hydrodata/re
## [1] "Successfully read - Lake Powell : Area - https://www.usbr.gov/uc/water/hydrodata/reservoir_da
## [1] "Successfully read - Lake Mead : Storage - https://www.usbr.gov/uc/water/hydrodata/reservoir_d

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Evaporation - https://www.usbr.gov/uc/water/hydrodata/res

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Inflow - https://www.usbr.gov/uc/water/hydrodata/reservoir

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Inflow Volume - https://www.usbr.gov/uc/water/hydrodata/r
```

```

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Unregulated Inflow - https://www.usbr.gov/uc/water/hydrodata/

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Unregulated Inflow Volume - https://www.usbr.gov/uc/water/
## [1] "Successfully read - Lake Mead : Total Release - https://www.usbr.gov/uc/water/hydrodata/reser
## [1] "Successfully read - Lake Mead : Release volume - https://www.usbr.gov/uc/water/hydrodata/rese
## [1] "Successfully read - Lake Mead : Pool Elevation - https://www.usbr.gov/uc/water/hydrodata/rese

## Error in file(file, "rt") : cannot open the connection
## [1] "Skipping due to error - Lake Mead : Area - https://www.usbr.gov/uc/water/hydrodata/reservoir_

## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/Ba
## [1] "C:/Rosenberg/Work/USU/Research/ColoradoRiver/RCode/ColoradoRiverCollaborate/AutoReadUSBRData/Re

```

Figure 1: Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

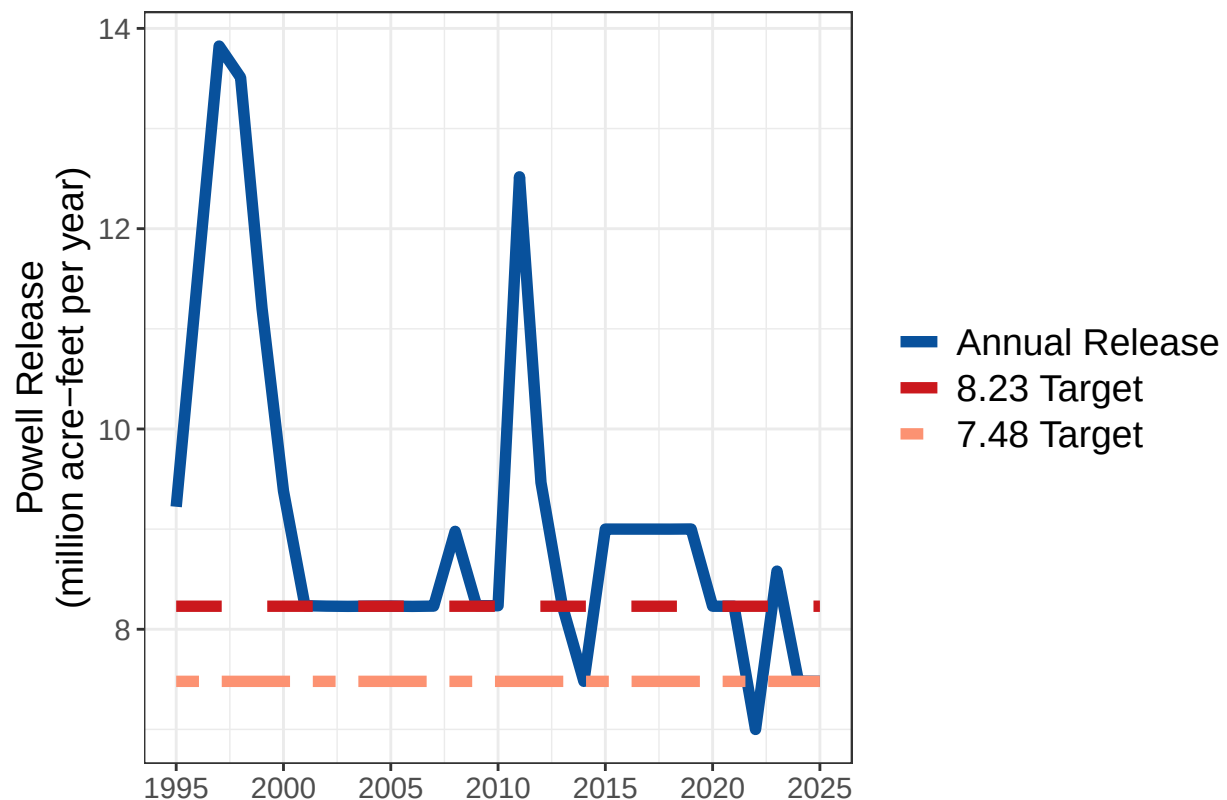


Figure 1: Ten-Year Powell Release Compared to 10-year Targets

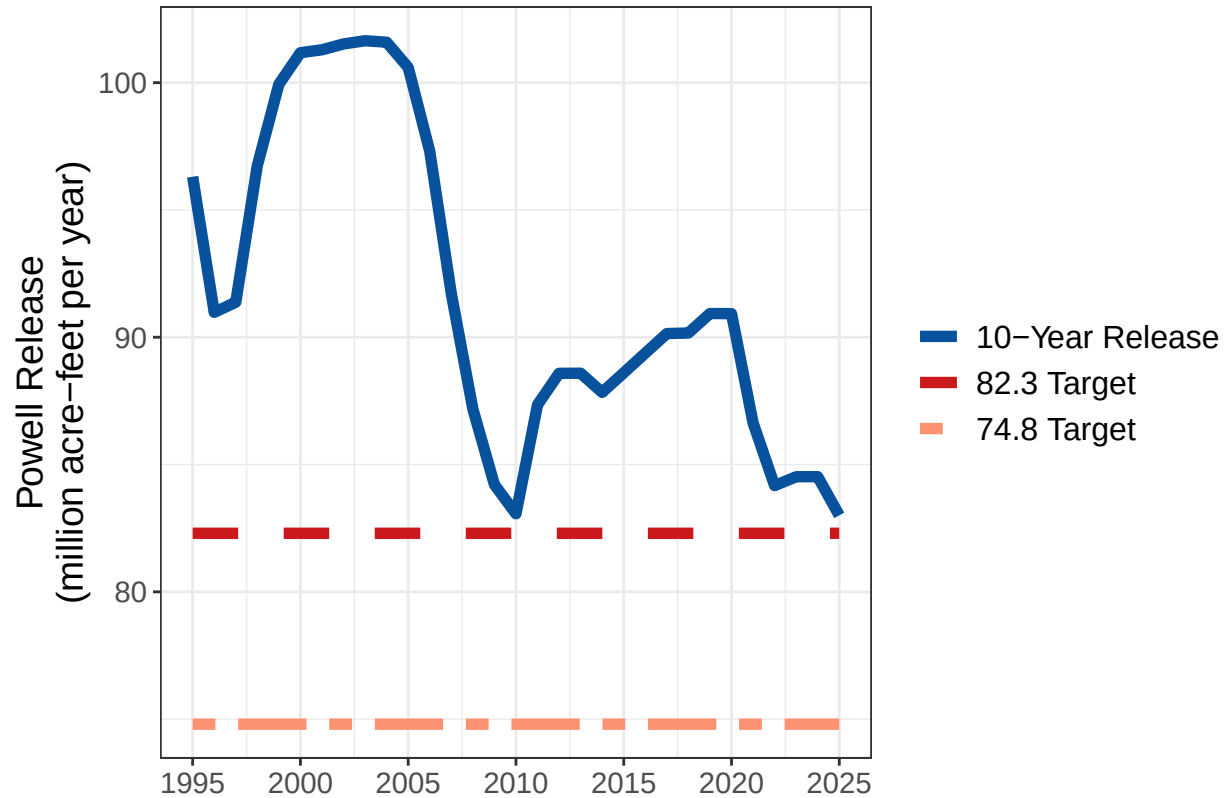


Table 1: Current Storage and Elevations

[1] "Reservoir Data on: May 17 , 2026"

A tibble: 2 x 3

Groups: ResName [2]

	ResName	Storage	Pool Elevation
	<chr>	<dbl>	<dbl>
1	Lake Powell	5.67	3527.
2	Lake Mead	7.86	1052.

Figure 3: Lake Powell, Lake Mead, and Combined Storage

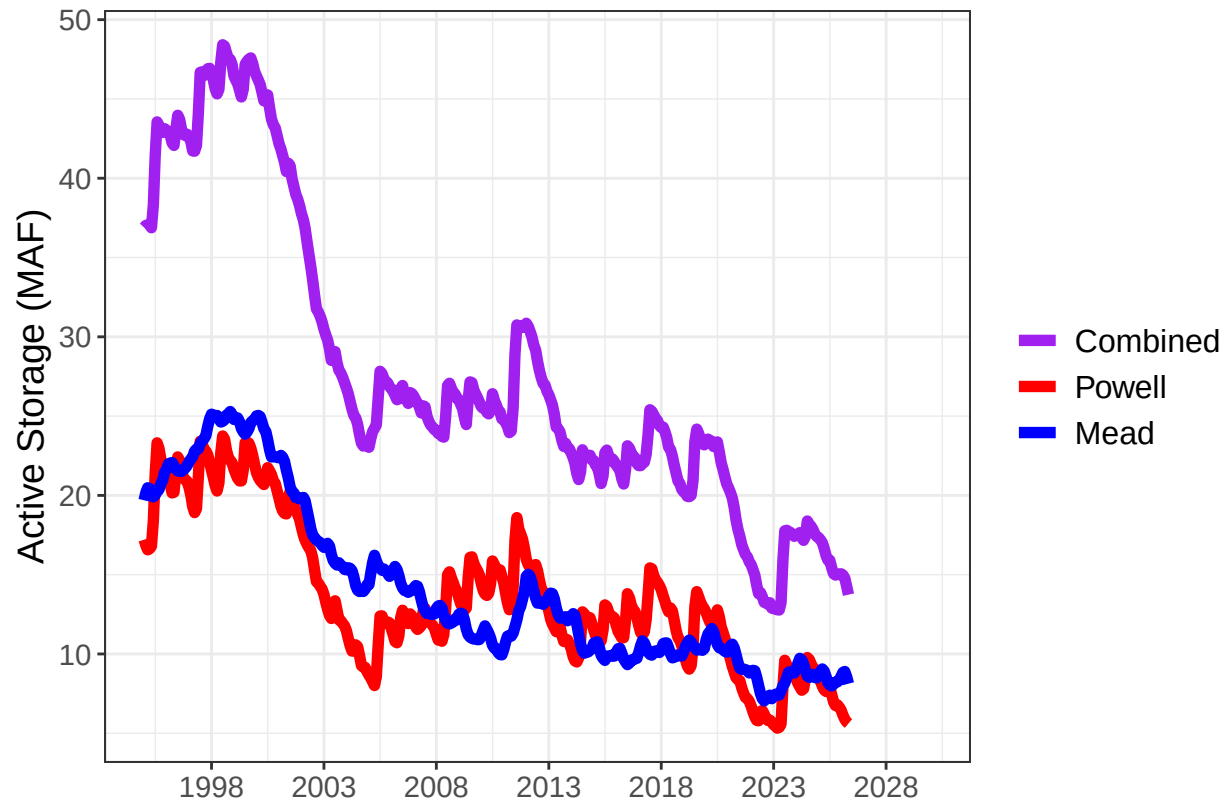


Figure 4: Lake Powell Storage vs Lake Mead Storage

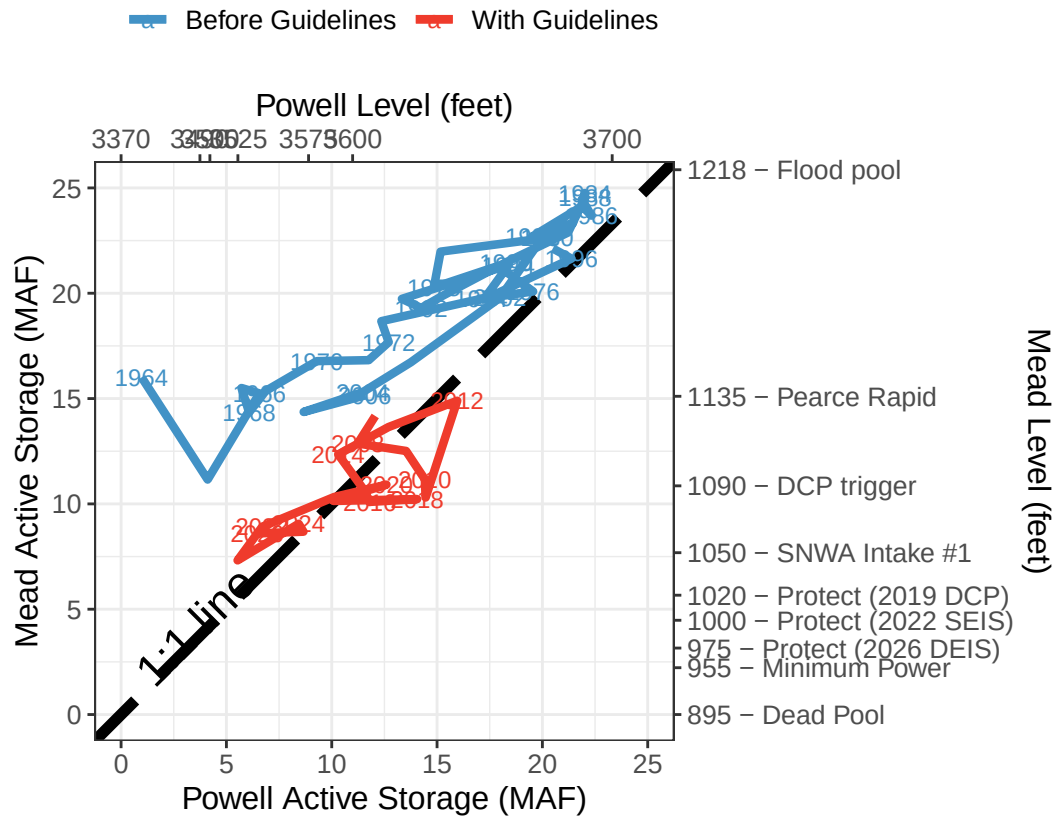


Figure 5: Lake Powell Storage and Elevation over Time.



Figure 6: Lake Mead Storage and Elevation over Time.

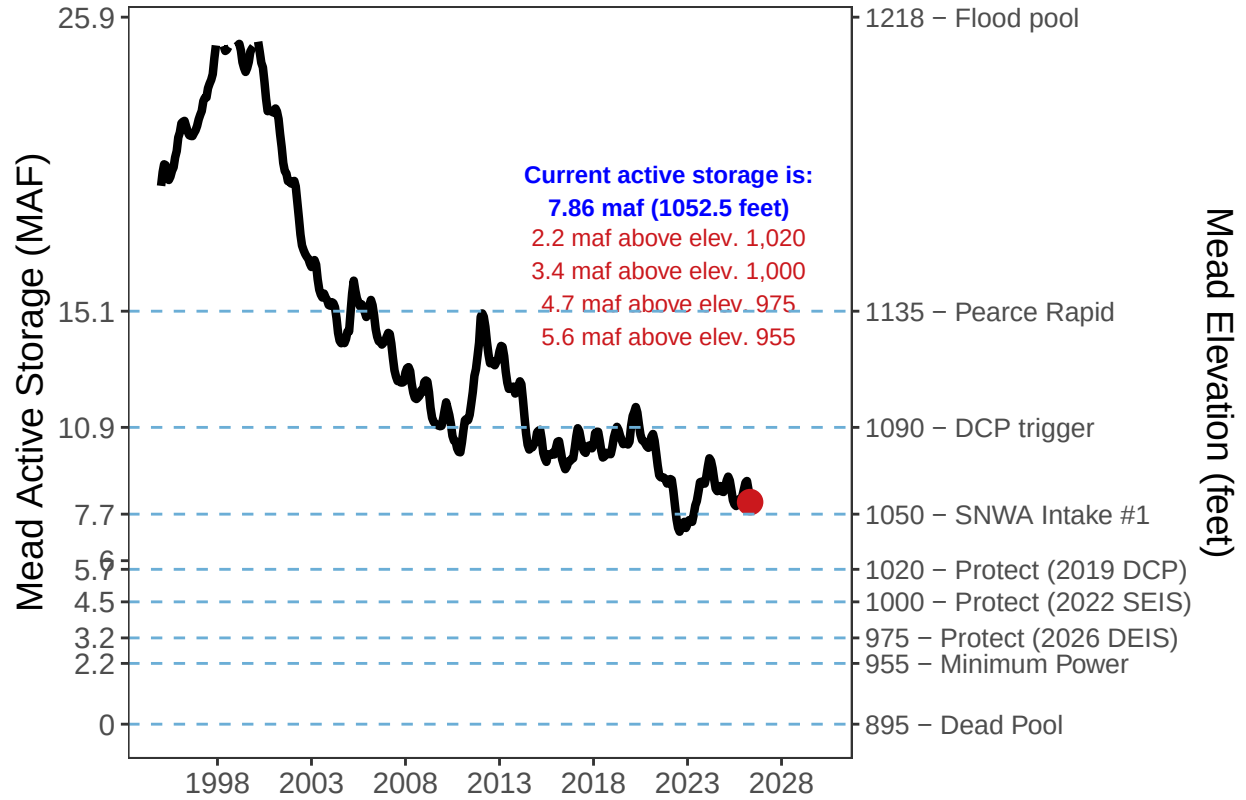


Figure 6B: Lake Powell Daily Inflow and Release

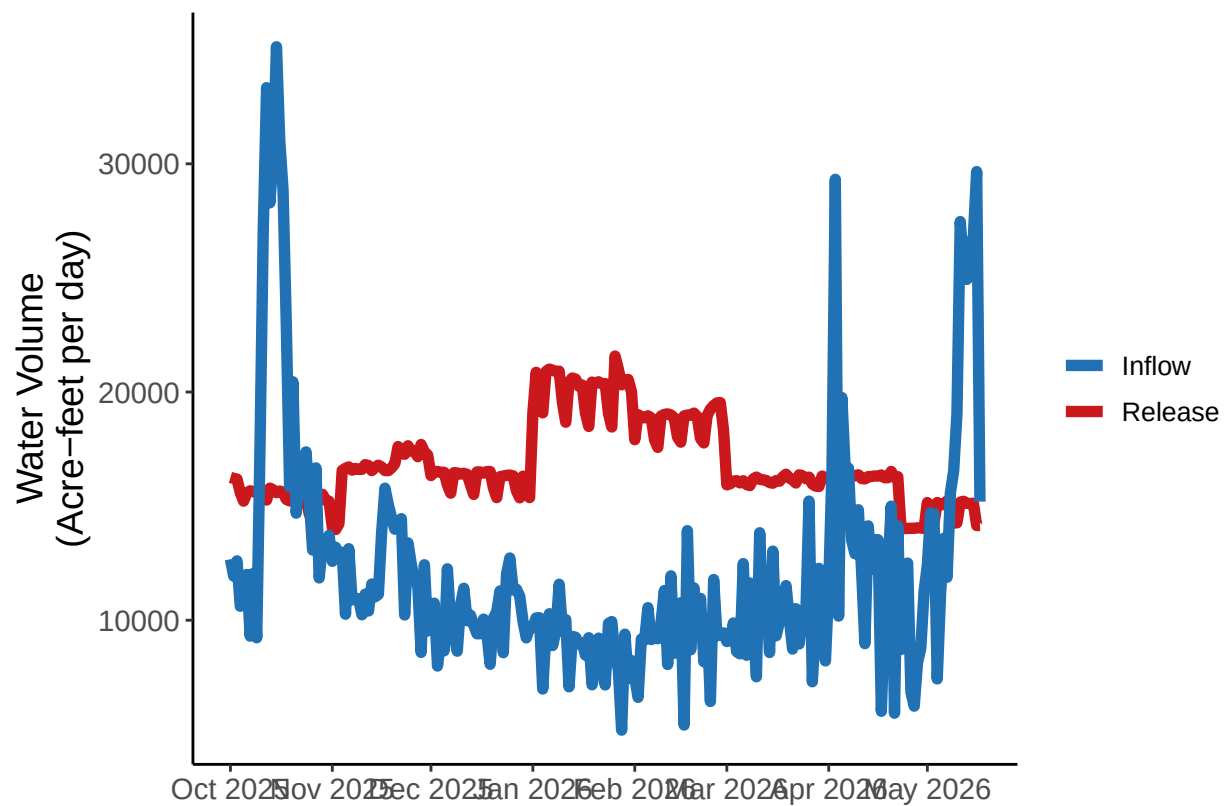


Figure 7: Frequency of Annual Lake Powell Release Volumes.

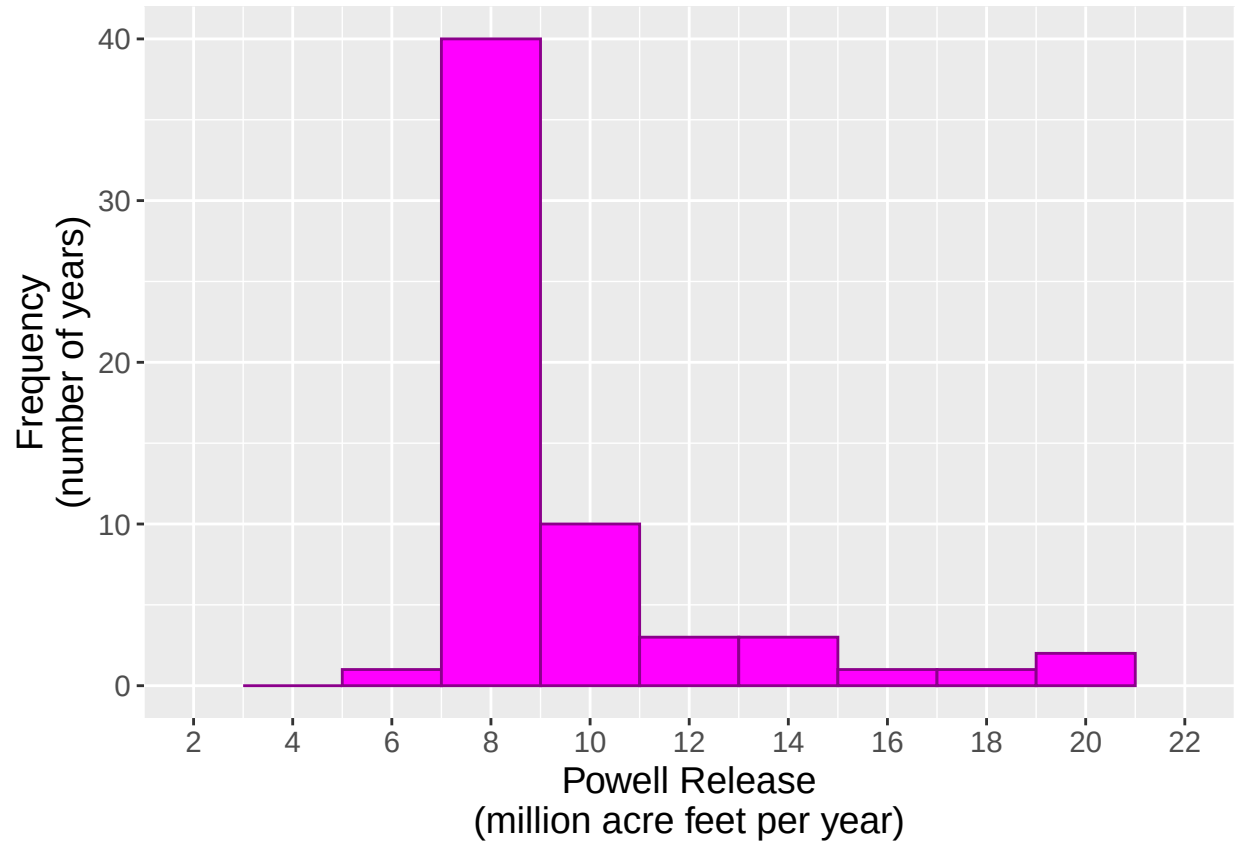


Figure 8: Frequency of Annual Lake Powell Inflow Volumes.

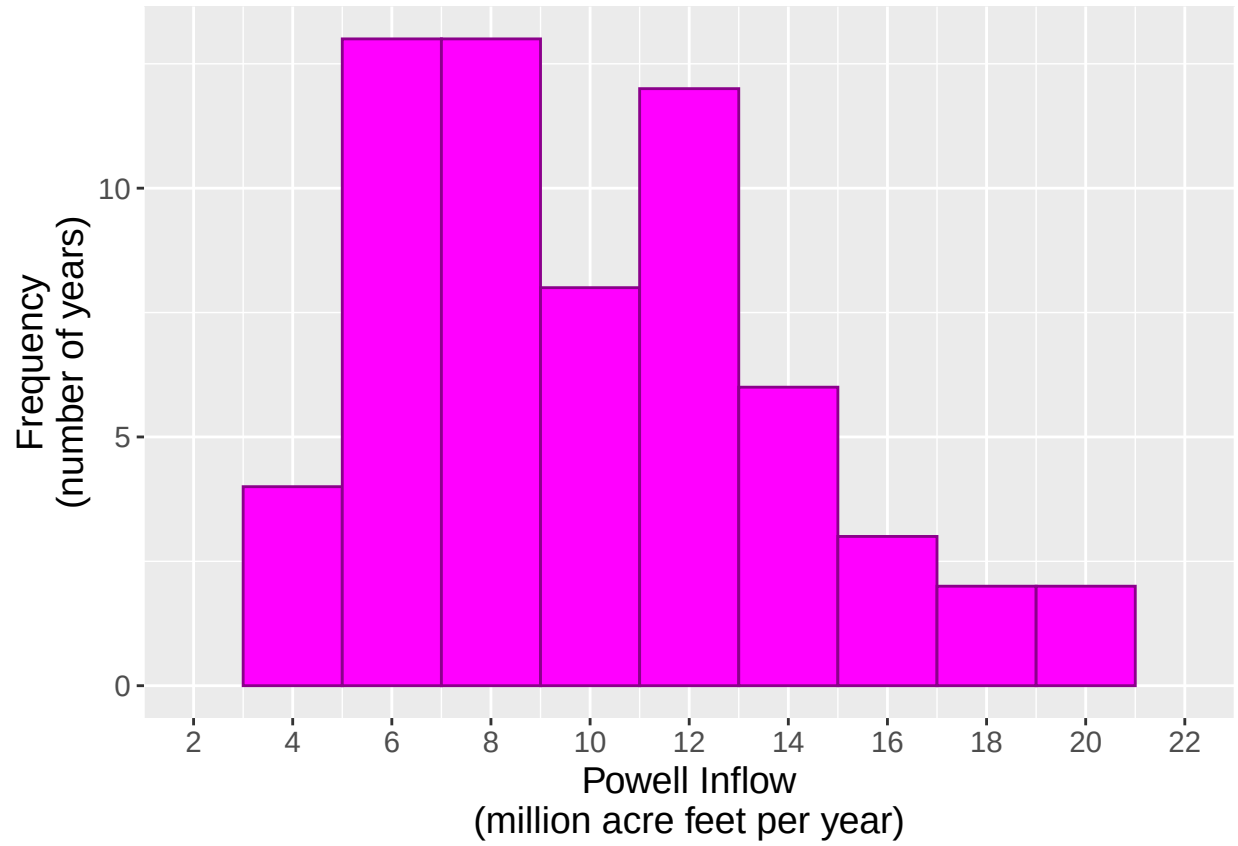


Figure 9: Lake Powell Inflows and Releases.

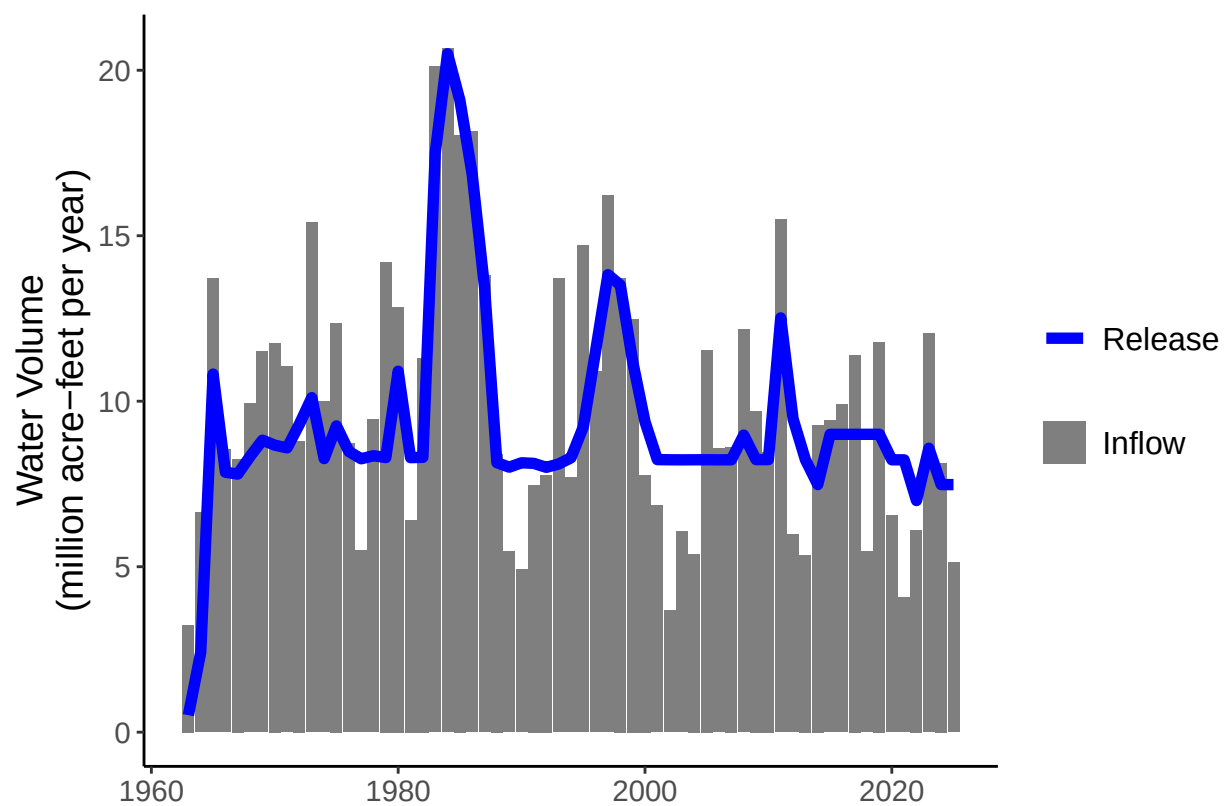


Figure 10: Annual Lake Powell Evaporation.

